

Research Interests

- Clouds and chemistry in brown dwarf and exoplanet atmospheres through spectroscopic observations and atmospheric retrievals
- Deep learning for atmospheric modelling and retrievals

Education

- 2024 – **PhD in Astronomy**, *University of Hertfordshire, UK*
- **Project:** *Remote Sensing Alien Clouds and Chemistry with JWST*
 - **Supervisors:** Dr. Ben Burningham, Dr. Fei Wang, Prof. Hugh Jones
- 2021 – 2023 **MSc in Astronomy Research**, *Leiden University, Netherlands, Grade: 7.5/10 (US: A-, UK: Upper Second Class)*
- **2nd MSc Thesis:** *Impact of Thermal Evolution on Exoplanetary Atmospheres*
 - **Supervisors:** Dr. Yamila Miguel, Amy Louca
 - **1st MSc Thesis:** *Substructure of Planet-Forming Disks with VLT*
 - **Supervisors:** Prof. Dr. Michiel Hogerheijde, Dr. József Varga
- 2020 – 2021 **MSc in Astronomy**, *Eötvös Loránd University, Hungary, Discontinued, went to Leiden instead*
- 2017 – 2020 **BSc in Earth Sciences**, *Eötvös Loránd University, Hungary, Grade: 5.0/5.0 (US: A+, UK: First Class)*
- **BSc Thesis:** *Lightcurves of Hilda Asteroids and Transneptunian Objects*
 - **Supervisor:** Dr. Csaba Kiss

Work Experience

- Nov 2024 – **Visiting Lecturer**, *Department of Physics, Astronomy and Maths, University of Hertfordshire, UK*
- Present
- Leading PC lab sessions, invigilating exams, and grading assignments for Data Science MSc students.
 - Data Handling and Visualisation: Fall 2026, Spring 2025
 - Machine Learning and Neural Networks: Spring 2026, Fall 2025
 - Applied Data Science 1: Spring 2026
- Sep 2022 – **Teaching Assistant**, *Leiden University, Faculty of Science, Netherlands*
- Dec 2022
- Led practical sessions and graded weekly assignments for the mandatory Astronomy BSc course, 'Stars'.
- Oct 2019 – **Observatory Assistant**, *Konkoly Observatory, HUN-REN Research Centre for Astronomy and Earth Sciences, Hungary*
- Aug 2021
- Performed observations at Piskéstető Mountain Station (3-4 nights every month).
 - Assisted in data reduction and analysis of the observed data.

Publications

First-authored

5. **Kecskeméthy, V.**; Burningham B., et al. (exp. submission in 2027) *Clouds, Chemistry, and Thermal Structure Across the L–T Sequence: Neural Emulator Atmospheric Retrievals of Twelve Benchmark Brown Dwarfs*. (In preparation).
4. **Kecskeméthy, V.**; Burningham B., et al. (exp. submission in 2026) *A Neural Network Forward Model for High-Dimensional Retrievals of Brown Dwarf Atmospheres*. (In preparation).
3. **Kecskeméthy, V.**; Burningham B., et al. (2026) *Benchmark Brown Dwarfs as Chemical Laboratories: Linking System Bulk Properties to Atmospheric Retrievals*. *Monthly Notices of the Royal Astronomical Society*, Volume 552, Issue 1, October 2026. [\[ADS\]](#)
2. **Kecskeméthy, V.**; Louca, A.; Miguel, Y. (2024). *Temporal Variability in Transmission Spectra of H₂-Dominated Exoplanets: The Influence of Thermal Evolution and Stellar Irradiation on Atmospheric Composition*. *New Astronomy*, Volume 113, December 2024. [\[ADS\]](#)
1. **Kecskeméthy, V.**; Kiss, C., et al. (2023). *Light Curves of Transneptunian Objects from the K2 Mission of the Kepler Space Telescope*. *The Astrophysical Journal Supplement Series*, Volume 264, Issue 1, January 2023. [\[ADS\]](#)

Co-authored

5. Rothermich, A.; [and 18 others, including **Kecskeméthy, V.**] (2026). *The Buddy System: Characterizing the benchmark system containing the T0 brown dwarf CWISE J210640.16+250729.0 using JWST.* (Accepted)
4. Varga, J.; [and 39 others, including **Kecskeméthy, V.**] (2024). *Mid-infrared Evidence for Iron-Rich Dust in the Multi-Ringed Inner Disk of HD 144432.* *Astronomy & Astrophysics*, Volume 681, January 2024. [ADS]
3. Csörnyei, G.; [and 9 others, including **Kecskeméthy, V.**] (2022). *Study of Changes in the Pulsation Period of 148 Galactic Cepheid Variables.* *Monthly Notices of the Royal Astronomical Society*, Volume 511, Issue 2, April 2022. [ADS]
2. Marciniak, A.; [and 67 others, including **Kecskeméthy, V.**] (2021). *Properties of Slowly Rotating Asteroids from Convex Inversion Thermophysical Model.* *Astronomy & Astrophysics*, Volume 654, October 2021. [ADS]
1. Kalup, C.; [and 10 others, including **Kecskeméthy, V.**] (2021). *101 Trojans: A Tale of Period Bimodality, Binaries, and Extremely Slow Rotators from K2 Photometry.* *The Astrophysical Journal Supplement Series*, Volume 254, Issue 1, May 2021. [ADS]

Talks & Posters

5. **Benchmark Brown Dwarfs as Chemical Laboratories** (Talk) Santa Cruz, CA, USA
Other Worlds Laboratory (OWL) 2026
4. **Silicate Clouds and Chemical Inheritance in Benchmark Brown Dwarfs** (Talk) Monterey, CA, USA
Charting Substellar Worlds 2026
3. **Investigating Silicate Clouds in Benchmark L-T Dwarfs with JWST** (Poster) Cork, Ireland
European Astronomical Society Annual Meeting 2025
2. **SDSSJ141659.78+500626.4: Investigating Silicate Clouds on an L dwarf** (Talk) Hatfield, UK
UH-PAM Lunch talk 2025
1. **NH₃ observations of the Orion Molecular Cloud** (Talk) Budapest, Hungary
Interaction of Stars with their Environment 2019

Scholarships, Awards & Grants

- Aug 2026 **Other Worlds Laboratory (OWL) Travel Grant**, UC Santa Cruz
- July 2026 **Charting Substellar Worlds Travel Grant**, UC Santa Cruz
- 2024 – 2028 **Science and Technology Facilities Council (STFC) PhD Studentship**, University of Hertfordshire
- Spring 2020 **Research Scholarship (tudományos ösztöndíj)**, Eötvös Loránd University
- Dec 2019 **Young Scientists' Conference (TDK)**: Submitted 2 projects, awarded 1st and 2nd place, Eötvös Loránd University
- Spring 2018 – **Academic Excellence Scholarship (tanulmányi ösztöndíj)**: Won it four times, Eötvös Loránd University
- Fall 2019

Software Development

- **Main developer of Brewster Emulator:** this is a work in progress, in which I am replacing Brewster's forward model with a neural network to accelerate retrieval analyses. Currently, the emulator is $\sim 400\times$ faster than the forward model.
- **Development contributions to Brewster v2:**
 - implemented universal instrument-resolution convolution using user-supplied resolving power files
 - integrated FastChem for gas phase chemistry and condensation calculations
 - added an easy-to-use cloud diagnostics for $\tau = 1$ surfaces, species masses, number densities, and Mg/Si ratios

Technical skills

- **Research methods:** Atmospheric retrievals, Bayesian inference, radiative transfer, low-resolution spectroscopy, cloud modelling, deep learning
- **Softwares:** Brewster, petitRADTRANS, emcee, FastChem, VULCAN
- **Programming languages:** Python, IDL, C, Fortran